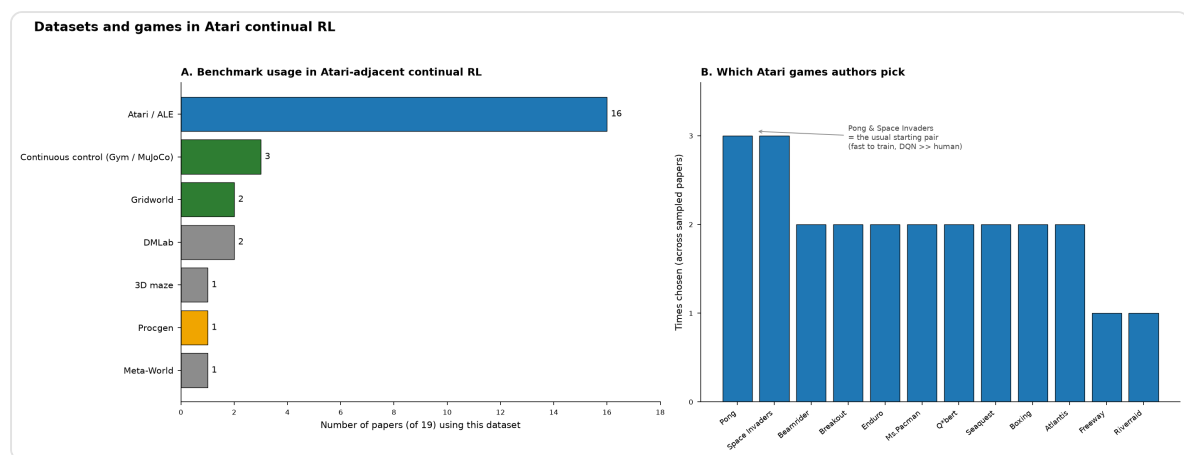


Continual Reinforcement Learning on Atari — Literature Summary

Scope: 19 papers on sequential / continual RL on Atari and adjacent benchmarks, mapped for a constrained local↔global (min-max) consolidation method. Prepared as a planning brief for the paper.

1. Datasets people use

Atari / ALE dominates (16 of 19 papers), but almost always as a hand-picked 3–10 game subset rather than the full 57-game suite. Continuous control (Gym / MuJoCo) and gridworlds are the common lighter-weight alternatives — and the sensible places to prototype before scaling to Atari.



A: benchmark usage across the 19 papers. **B:** most frequently chosen Atari games.

Benchmark	Papers	Role for us
Atari / ALE	16	Headline benchmark; use a 3–6 game subset
Continuous control (Gym / MuJoCo)	3	prototype here natural home for a policy-constraint method (TRPO/PPO)
Gridworld / MiniGrid	2	prototype here fastest iteration, trivial task boundaries
DMLab	2	3D navigation, cyclic task sequences
3D maze	1	—
Procgen	1	Procedural generalization / plasticity tests
Meta-World	1	Modern robotic-manipulation CRL benchmark

2. Games most frequently used

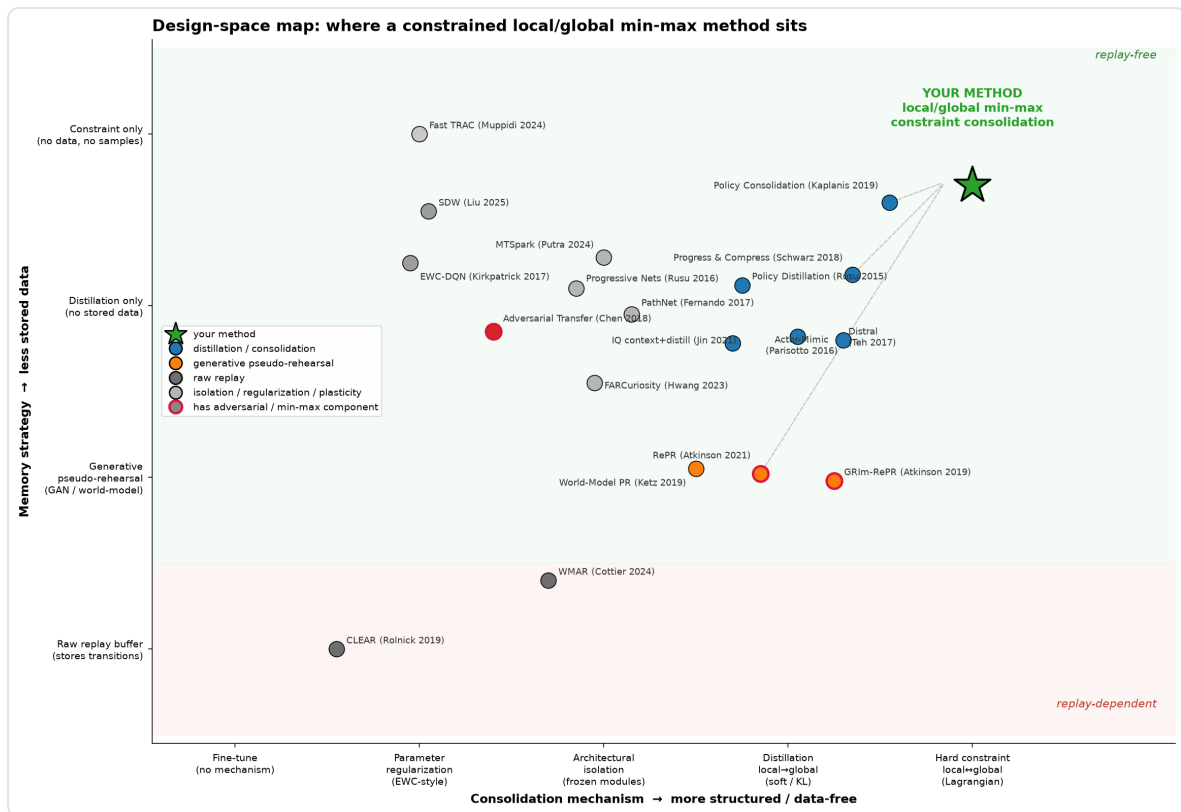
Pong and **Space Invaders** are the most common picks (each in 3 papers), followed by a stable second tier. Authors deliberately choose games where DQN is already *far above human*, so a clean “stays above human throughout the sequence” claim is possible.

Frequency	Games
3×	Pong, Space Invaders
2×	Beamrider, Breakout, Enduro, Ms. Pacman, Q*bert, Seaquest, Boxing, Atlantis
1×	Freeway, Riverraid, Krull, Hero, Stargunner, Road Runner, James Bond, Crazy Climber

Recommendation: a 3–6 game sequence anchored on **Pong** → **Boxing** → **a third**, matching RePR's scale so results are directly comparable. Consider cyclic revisits (Progress & Compress protocol) as the harder retention test reviewers expect.

3. Approach landscape & where our method sits

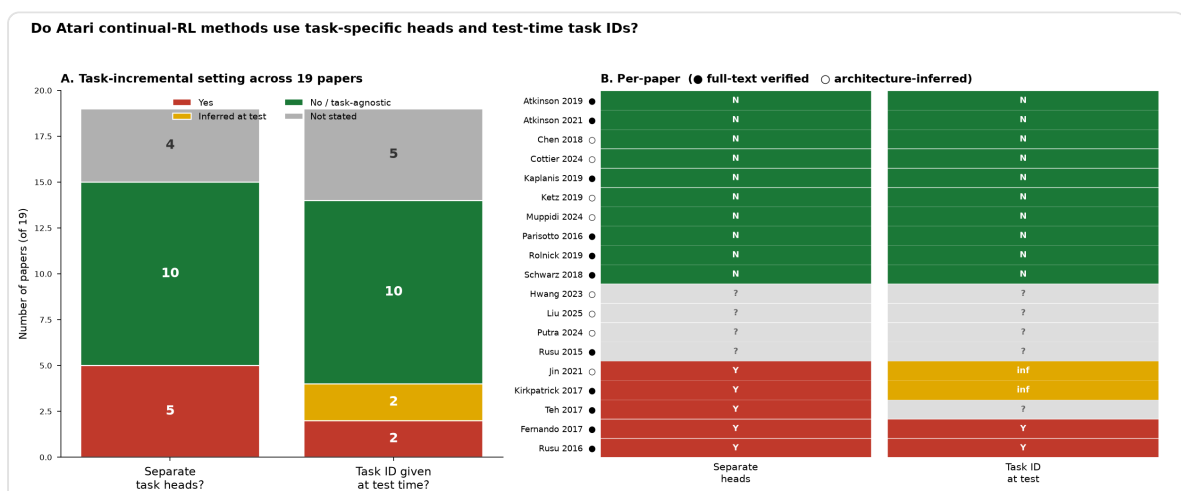
Seven method families; **most Atari CRL methods are already replay-free** (17 of 19), achieving retention through distillation, generative pseudo-rehearsal, or regularization rather than a raw buffer. A **hard Lagrangian constraint enforced symmetrically between two alternating policies** is the one unoccupied corner — our contribution.



Design-space map: consolidation mechanism (x) × memory strategy (y). Our method (green star) occupies the empty hard-constraint / no-data corner; nearest neighbours are Progress & Compress, Policy Consolidation, and RePR (one axis away each).

4. Task-specific heads & test-time task identifiers

A key architectural axis often left implicit: does the method use **separate output heads/columns per task**, and is it **given the task identifier at test time**? Methods that need the task ID at inference solve the easier *task-incremental* problem; task-agnostic methods (no ID at test) are the harder and more general setting — and the one our alternating local↔global scheme should target. Of the 19 papers (11 verified from full text, 8 from architecture):



A: counts across 19 papers. **B:** per-paper grid (● full-text verified, ○ architecture-inferred). Y = yes, N = no / task-agnostic, inf = task ID inferred at test, ? = not stated.

Question	Yes	No / task-agnostic	Inferred at test	Not stated
Separate task-specific heads?	5	10	—	4
Task ID provided at test time?	2	10	2	5

Takeaways:

- **Separate heads (5):** Progressive Nets, PathNet, Distral, EWC-DQN, IQ — the parameter-isolation and per-task-column methods.
- **No heads / single shared network (10):** incl. Progress & Compress (explicitly “no task-specific parameters”), RePR, Policy Consolidation, CLEAR, Actor-Mimic — the consolidation and pseudo-rehearsal families our method belongs to.
- **Task ID at test time:** only 2 *require* it (Progressive Nets, PathNet); EWC-DQN and IQ *infer* it via a task-recognition/context module (EWC reports only a modest gain from an oracle label); 10 need no ID; 5 don’t state it. (Distral uses per-task policies but states no explicit test-time task-ID protocol.)
- **For our paper:** the strongest positioning is **task-agnostic at test** (no ID, single network) — matching RePR / P&C / Policy Consolidation. State this setting explicitly, since only ~half the field does.

5. Figures & tables to have in our paper

Concrete deck for the paper, in recommended order. **MUST** = core to the story; **OPT** = strengthens if space allows.

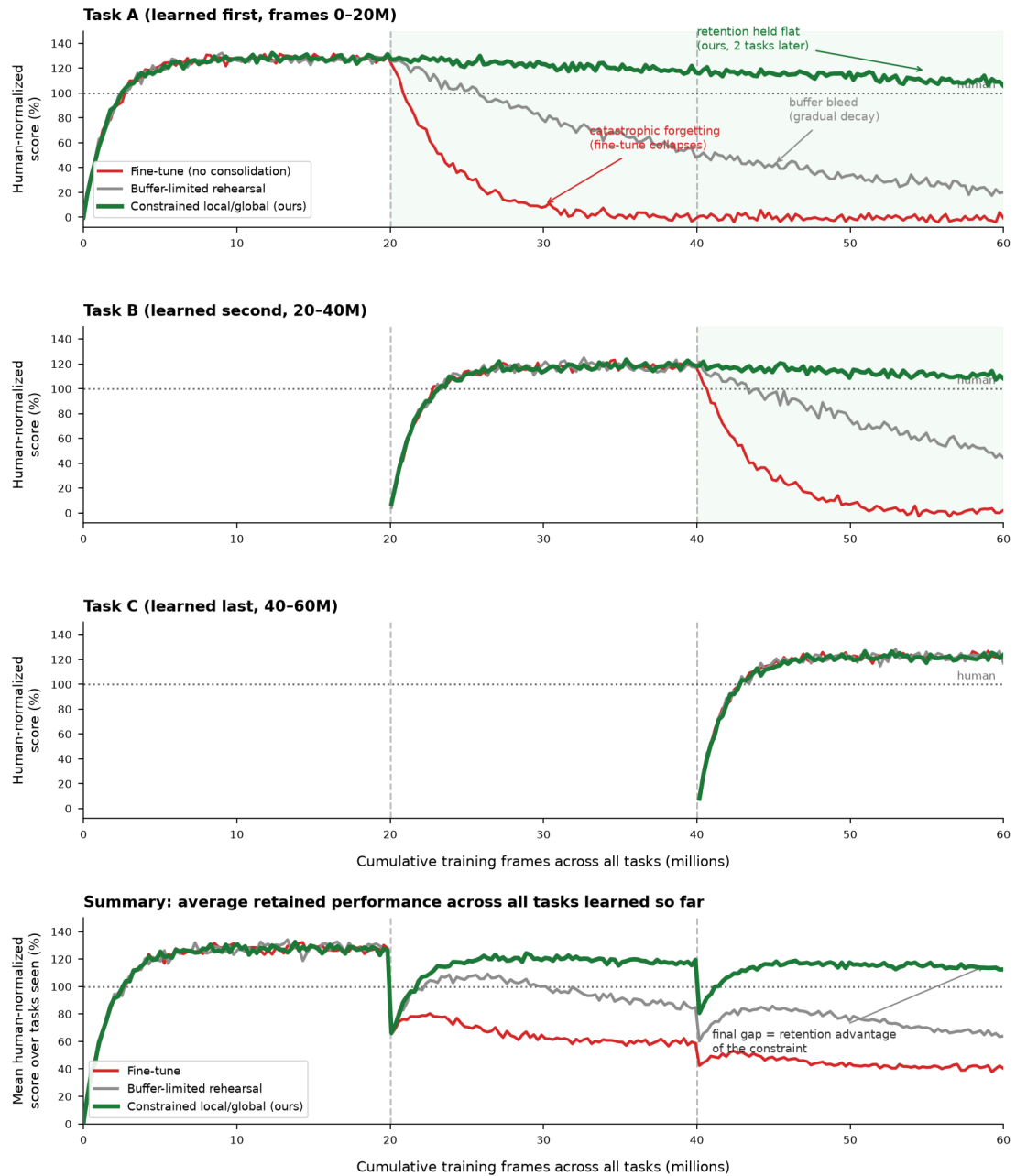
#	Item	What it shows	Priority
Fig 1	Design-space / positioning map	The gap our method fills vs. RePR, P&C, Policy Consolidation. Makes the novelty legible in one glance.	MUST
Fig 2	Method schematic	Local policy ↔ global policy alternation with the Lagrangian trust-region constraint (the min-max loop).	MUST
Fig 3	Per-task retention learning curves + summary panel	The core result: our curve flat in the retention region while a fine-tune baseline collapses and a buffer-limited baseline bleeds. Human-normalized axis; CI bands over seeds.	MUST
Tab 1	Final retention table	% of single-task expert retained after the full sequence, per method — a metric almost nobody reports cleanly.	MUST

Tab 2	Setup / protocol table	Games, sequence, #frames, revisits, buffer budget, params — for reproducibility & fair comparison.	MUST
Fig 3b	Constraint-satisfaction diagnostic	Per-task forgetting F_k vs. the ε bound over training, plus constraint-violation rate and Lagrange multiplier λ . Proves the hard constraint actually held.	MUST
Fig 4	Buffer-budget / ablation sweep	Retention vs. replay-buffer size, showing the buffer alternative fails regardless of budget; and constraint-strength ablation.	OPT
Fig 5	Task-order robustness	Variance across task orderings — ties directly to your prior HTCL work; a differentiator few Atari papers show.	OPT

Experimental check — constraint satisfaction. Because the method is defined by a hard forgetting constraint, every run must **verify the constraint is actually satisfied**, not just that mean reward looks healthy. At each consolidation step (and at every evaluation checkpoint), log the per-task forgetting $F_k = J_k^{\text{best}} - J_k^{\text{current}}$ for each past task k and assert $F_k < \varepsilon$ (equivalently, the KL / trust-region divergence between local and global on past tasks stays within the bound). Report this as: (i) a **constraint-violation rate** (fraction of checkpoints or tasks where $F_k \geq \varepsilon$), and (ii) the **active Lagrange multiplier λ** over training — a multiplier pinned at its ceiling flags an infeasible ε . A method that hits its reward target while silently violating $F_k < \varepsilon$ has not actually satisfied its own formulation, so this check is a required panel/column, not an optional diagnostic.

Template for Fig 3 (drop in real results)

TEMPLATE (synthetic data) — continual-RL retention plot for a constrained local/global method

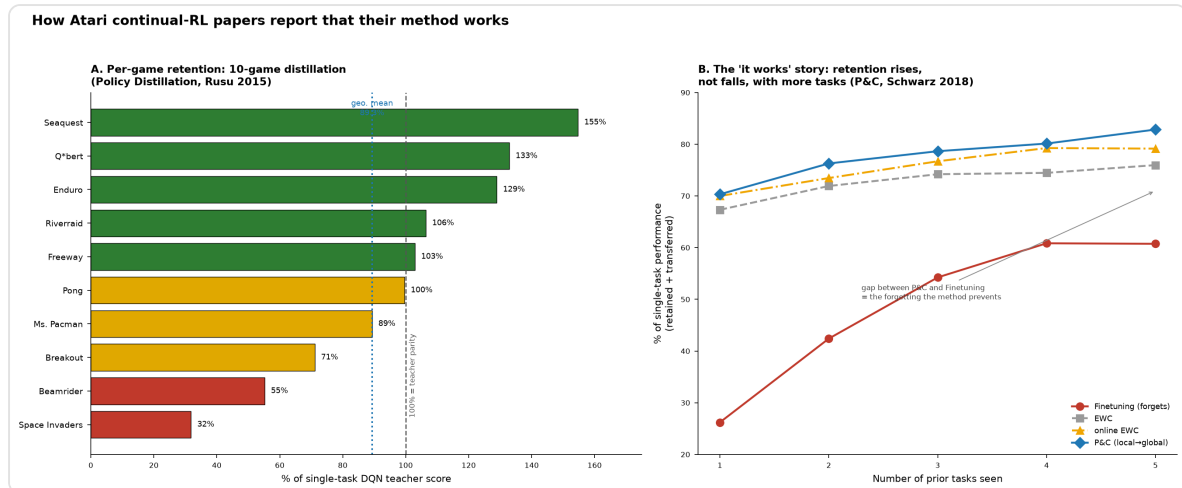


Illustrative synthetic curves showing the intended layout and story — replace with measured results. Green band = retention region (task no longer trained).

Template (synthetic) in the field-standard layout: one panel per task + a mean-retention summary. x = cumulative frames; dashed lines = task switches; green band = retention region (task no longer trained).

Replace synthetic curves with logged, human-normalized scores and add CI bands.

Reference examples from the literature



Left: the per-game retention table style (Policy Distillation, 89.3% geo-mean of teachers). Right: the “retention rises with more tasks” story (Progress & Compress, Schwarz 2018). These are the two report styles our Tab 1 / Fig 3 should beat.

Key reporting gaps to exploit:

- **Retention-as-%-of-expert** after the full sequence — reported cleanly by only 2 of 19 papers.
- **Protect the critic**, not just the policy — existing methods leave the value function to forget.
- Use **stratified-bootstrap confidence intervals** (reliable protocol) given small-N deep-RL noise.

Supporting data: `atari_crl_dataset.csv` (19-paper structured dataset),
`policy_distillation_pergame.csv`, `progress_compress_transfer.csv`.